

Metagenomic Taxonomic Classifiers

Comparing marker-gene and whole-genome database approaches

Learning Objectives

- Understand the goals of metagenomic taxonomic classification
- Compare marker-gene and whole-genome database approaches
- Recognize strengths and limitations of popular classifiers
- Choose appropriate tools for different study designs

What is Taxonomic Classification?

- Assigns sequencing reads (or contigs) to taxa
- Used for community profiling in microbiomes and environmental samples
- Challenges include short reads, unknown organisms, and database bias

Two Major Strategies

- Marker-gene based classifiers
- Whole-genome / whole-database classifiers
- Main tradeoff: speed & precision vs sensitivity & completeness

Marker-Gene Approaches

- Use a curated set of phylogenetically informative genes
- Examples: MetaPhlAn, mOTUs, PhyloSift, sylph
- Reads mapped only to marker genes (unique single copy marker genes)
- Typically smaller databases and faster runtimes

Whole-Genome Database Approaches

- Compare reads against all available genomes or protein databases
- Examples: Kraken2, Centrifuge, Kaiju, CLARK
- Use k-mers, exact matching, or translated alignment
- Typically higher sensitivity but larger databases

Popular Marker-Gene Classifiers

- MetaPhlAn: clade-specific marker genes
- mOTUs: universal single-copy marker genes
- PhyloSift: phylogenetic marker genes with tree placement
- Often optimized for accurate relative abundance estimation

Popular Whole-Genome Classifiers

- Kraken2: exact k-mer matching
- Centrifuge: compressed indexing for low memory use
- Kaiju: protein-level translated search
- CLARK: discriminative k-mers

How Kraken2 Works

- Builds k-mer database from reference genomes
- Each k-mer linked to lowest common ancestor (LCA)
- Reads classified by majority vote of matching k-mers
- Very fast and scalable

How MetaPhlAn Works

- Uses unique clade-specific marker genes
- Reads mapped only to curated markers
- Estimates organismal abundance from marker coverage
- Lower false positives for known taxa

Marker-Gene vs Whole-Database

- Marker genes: higher precision, lower sensitivity
- Whole genome databases: higher sensitivity, more false positives (they use also informative sections of the genome)
- Marker methods require conserved informative genes
- Whole-genome methods better for strain-level analysis

Database Size and Computational Cost

- Marker-gene databases are compact
- Whole-genome databases may require >100 GB
- Kraken2 and Centrifuge optimized for speed
- Protein-level tools like Kaiju are slower but detect distant taxa

Performance Tradeoffs

- Precision: avoiding false assignments
- Recall/Sensitivity: detecting low abundance organisms
- Runtime and memory requirements vary greatly
- Database completeness strongly influences results

When to Use Marker-Gene Methods

- Rapid microbiome profiling
- Human gut and well-characterized environments
- Low computational resources
- Studies prioritizing robust abundance estimates

When to Use Whole-Database Methods

- Discovery of diverse or novel organisms
- Environmental metagenomics
- Strain-level characterization
- Detection of viruses or rare taxa

Important Caveats

- Reference database bias affects all methods
- Unknown organisms may remain unclassified
- Read length and sequencing depth matter
- Combining methods often improves confidence

Recommended Workflow

- Quality control of reads
- Initial rapid profiling with MetaPhlAn or Kraken2
- Validation using complementary methods
- Functional analysis after taxonomic profiling

Summary

- Marker-gene methods focus on informative conserved genes
- Whole-genome methods use broader reference databases
- No single classifier is best for all studies
- Tool choice depends on accuracy, speed, and study goals

Classifier Comparison Table

Tool	Approach	Main Method	Strength	Weakness
MetaPhlAn	Marker genes	Marker alignment	High precision	Misses novel taxa
Kraken2	Whole genome	k-mer matching	Very fast	Large databases, false positives?
Centrifuge	Whole genome	Compressed index	Low memory	Lower precision, false positives?
Kaiju	Whole genome	Protein search	Detects distant taxa	Slower
mOTUs	Marker genes	Single-copy markers	Good abundance estimates	Limited coverage